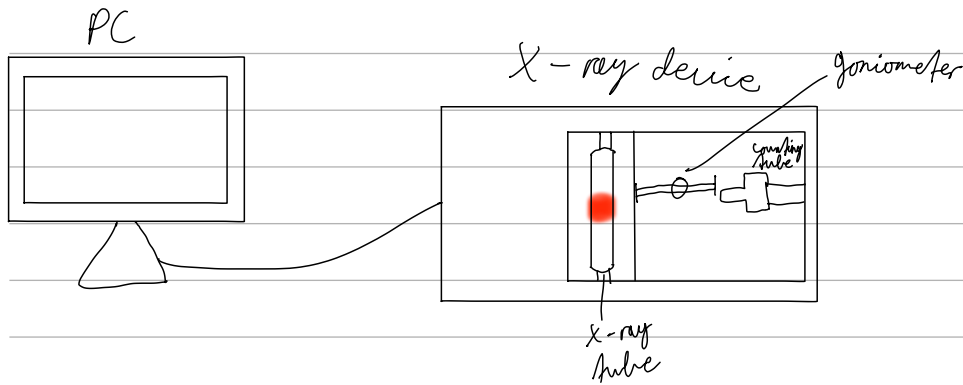


Equipment

- X-ray device with X-ray tube (Mo anode)
- Goniometer
- counting tube
- LiF- and NaCl crystal
- computer
- fluorescent screen with CCD-camera

SetupTask 1: X-ray spectrum of the LiF-crystal

We aim to measure the X-ray spectrum of our Mo anode with the LiF crystal. We set the device to $U = (35 \pm 0,1) \text{ kV}$, $I = (10 \pm 0,01) \text{ mA}$, $t = 5 \text{ s}$ and $\Delta B = 0,2^\circ$. We measure values between 3° and 23° . We save our data digitally.

Task 2: K_{α} and K_{β} lines of Mo

From the previous measurement, we select the locations of the K_{α} and K_{β} lines:

1. order: K_{α} at $10,2^\circ$, K_{β} at $9,0^\circ$
2. order: K_{α} at $20,6^\circ$, K_{β} at $18,2^\circ$

We repeat the measurements of the first task with $\Delta B = 0,1^\circ$ and $t = 20 \text{ s}$.

Task 3: counting rate as a function of V

We use an angle of $\beta = 7.5^\circ$ and measuring time of $t = 20\text{s}$. We measure the counting rate for values from 20 kV to 35 kV in steps of 1 kV

Table 1: $r(V)$

Voltage $V [\text{kV}]$	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
Rate $r [\frac{1}{\text{s}}]$ (average over 20s)	3	9,5	6,75	10,5	41,9	122,0	183,9	237,0	296,9	338,7	393,6	440,5	483,4	521,4	573,1	609,1

Task 4: Measurement of the x-ray spectrum with a NaCl crystal

We swap out the LiF crystal for a NaCl crystal, and measure the x-ray spectrum from 3° to 18° with the same options as Task 1. We again save our data digitally.

